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REVIEW ARTICLE

Regolith sintering and 3D printing for lunar construction: An extensive review on recent progress

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Abstract
In-Situ Resource Utilisation (ISRU) is increasingly being seen as a viable and essential approach to constructing infrastructure for human habitation on the moon. Transporting materials and resources, from Earth to the Moon, is prohibitively expensive and not sustainable for long-term, large-scale development. Various fabrication technologies have been investigated in recent years, designed for extra-terrestrial exploration and settlement. This review presents a comprehensive study on the development of several sintering techniques of lunar regolith simulant to demonstrate its feasibility for ISRU on the moon. Various critical processing parameters are evaluated in pursuit of creating a structural material that can withstand the extreme lunar environment. Key outcomes are summarised and assessed to provide insight into their viability. Finally, current challenges are addressed and potential improvements, and avenues for further research, suggested.

Keywords Lunar regolith · Space · Additive manufacturing · ISRU · Regolith sintering

Abbreviations

AM	Additive manufacturing
DIW	Direck ink writing
DLR	German aerospace centre
ESA	European space agency
ISRU	In-situ resource utilisation
KREEP	Rare-earth elements containing potassium and phosphorus
LEO	Low earth orbit
NASA	The national aeronautics and space administration
PSI	Physical Sciences Inc.
SPA	South Pole-Aitken
SPS	Spark plasma sintering

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1 Background of lunar construction

Recently, national space organisations and aerospace companies have begun working towards exploring the possibility of building a safe and secure human habitat on the Moon. One viable location is the Moon's South Pole, which has been thoroughly investigated over the years and contains significant ice deposits within its craters. These craters receive no sunlight and reach temperatures as low as -248 °C while other parts of the South Pole are exposed to sunlight for extended periods of time where the surface temperature ranges from approximately – 70 °C to 77 °C [1]. In addition to extreme temperature fluctuations, the lunar environment is exposed to solar and cosmic radiation, micrometeoroid impacts, and lunar dust [2]. A lunar base must shield humans from these conditions by providing appropriate infrastructure.

Current technology requires that all lunar facilities, equipment, and consumable materials must be transported from Earth. NASA's space shuttle could deliver payloads, of up to 27,500 kg, to low Earth orbit (LEO) at a cost of approximately $62 k/kg [3]. The current SpaceX Falcon 9 reusable two-stage rocket has reduced this cost to approximately $3 k/kg, with a payload capacity of 22,800 kg [4]. However, even with reusable launch systems, transporting the volume of material required to construct infrastructure on the lunar surface is likely to remain prohibitively expensive for the

foreseeable future. Therefore, the development of a sustainable long-term lunar base can only be realised through in-situ resource utilisation (ISRU), a concept to utilise locally resourced materials (lunar soil) to construct lunar infrastructure such as habitable lunar bases and landing pads [5].

Lunar soil, also known as regolith, is a raw material that envelopes the lunar surface and is the product of different processes comprised of continuous impact of large and micro-meteoroids, and bombardment of atomic particles from the Sun [2]. The characteristics of lunar regolith depend on the geography and history of its region. The lunar terrain is divided into three categories: procellarium KREEP terrane, feldspathic highlands terrane and South Pole-Aitken (SPA) Basin terrane [6]. A large amount of material data on lunar soil was collected by the Apollo and Luna programs in the early 70 s. Successive landings and remote observation missions, such as the Chang'E program, furthered understanding of the lunar surface.

Information on the specific geography, geology and nature of previous lunar landing sites has been outlined in other works [2, 7]. Several missions to the lunar surface are planned for the next decade, particularly aimed at the SPA region. There have been extensive remote studies of the SPA region, summarised by Moriarty et al. [8]. The South Pole is located in a highland region, bordering a unique pyroxene-rich band in the Aitken Basin. These studies and missions form the foundation for engineering studies using lunar simulants.

Research programs have been conducted to optimise utilisation of regolith for constructing building elements [9]. Sintering is a method to consolidate a loose powder into a solid material and is widely used in ceramics. The process involves high temperatures to initiate densification forming bonds between particles and are heated using various techniques including radiant heat and microwave radiation. Sintering is influenced by factors such as gas atmosphere, temperature gradient and pressure [10, 11]. Hence, sintering lunar soil could be the solution to meeting the structural requirements of a lunar base.

Figure 1 illustrates a publication trend of sintering techniques conducted on regolith simulants which comprise of publications on different sintering techniques from space organisations and researchers around the globe over the past decades. These publications were collected via the Web of Science Core Collection, an integrated base of scientific research. Boolean operators and manual screening were

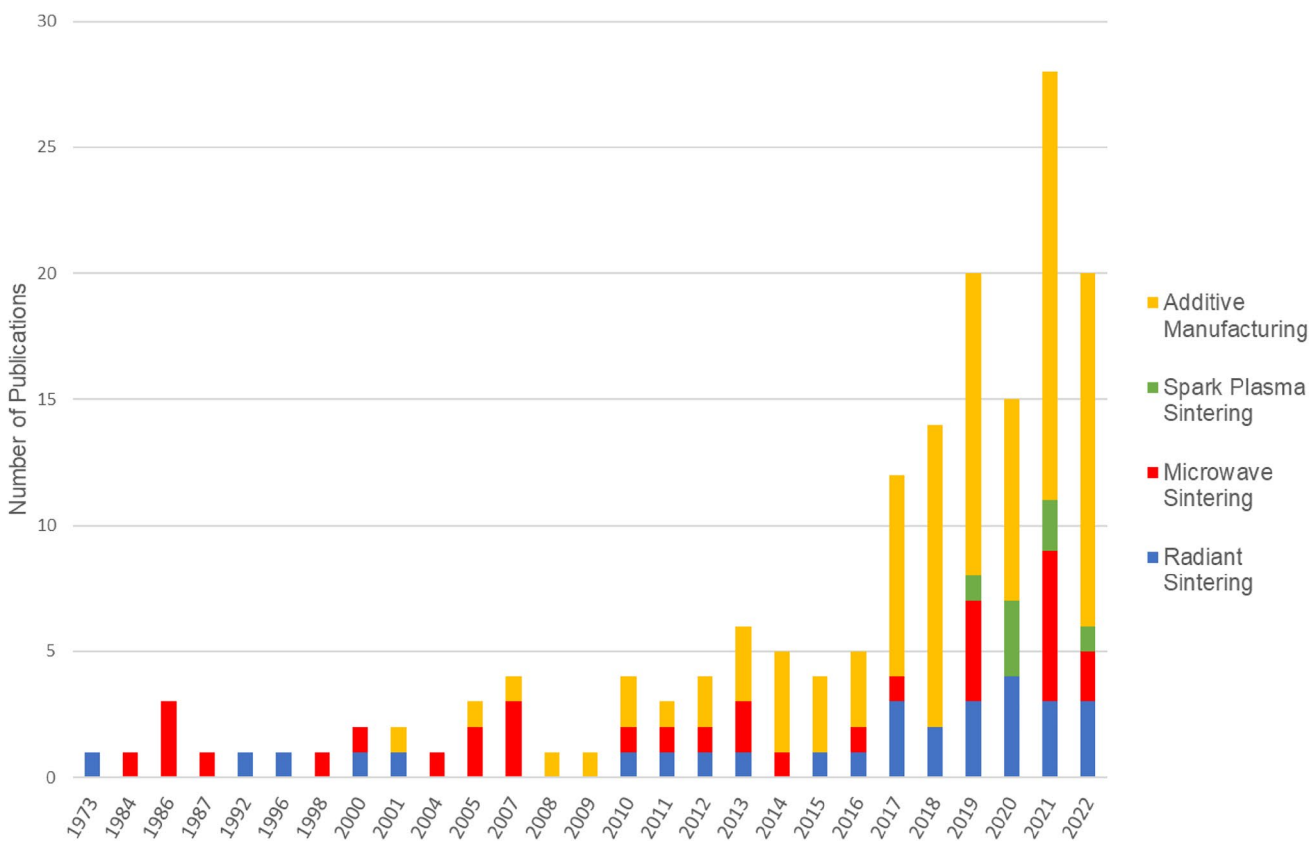

Fig. 1 Publication trends of sintering regolith throughout the decades collected via web of science core collection and refined through using Boolean operators and manual screening

applied to refine publications relative to the review paper. Publications relative to regolith sintering are interdisciplinary field whereas material manufacturing technology and mechanical characterisation are often integrated with each other which are hard to isolate to an individual field. During the first few decades, research on regolith sintering was limited particularly on additive manufacturing. However, since 2017, there has been a significant increase in experimental regolith sintering trials. This was driven by a renewed interest in the US, and other spacefaring nations, to return to the moon with the purpose of establishing a permanent human presence.

This paper highlights different sintering techniques for processing regolith simulants into a structural material. In Sect. 2, the origins and characteristics of lunar soil and regolith simulants are outlined and discussed in detail. Sintering methods in this review are divided into 'bulk' and 'focused' categories. Section 3 provides an overview of three bulk sintering approaches which include radiant sintering, microwave sintering and spark plasma sintering. This section will discuss key parameters and their influences on the mechanical properties (Density, hardness, porosity). Section 4 presents two locally focused sintering methods namely, selective laser sintering and solar sintering, and provides a discussion on the sintering behaviour. In Sect. 5, additive manufacturing (AM) applications using previous sintering methods are elaborated along with potential applications and challenges. This revisits selective laser sintering, solar sintering and microwave sintering, and examines Direct-Ink-Writing (DIW) as a point of comparison. With AM, the characteristics of each process and their general viability for different lunar industrial applications are explored. Finally, Sect. 6 concludes with key findings and directions for future work.

2 Regolith simulants and soils: classification and characterisation

The science retrieved by historical lunar landings and remote sensing missions has been consolidated, and the general lunar soil characteristics have been identified [2, 6, 12, 13]. These landings have largely laid the foundation of lunar studies in the following decades. The chemical composition and geotechnical characteristics of lunar soil are key to understanding its sintering behaviour. For Apollo samples, the soil composition was initially identified by petrographic 'particle counting' and spectral chemical studies [14]. Successive works in the 2010s examined the chemical composition and surface characteristics of lunar soil in detail [14–16]. An overview of lunar landing sites is presented in Fig. 2, and the details of numerous sites are tabulated in Table 1.

Due to the limited access to lunar samples on Earth, materials that simulate lunar soils (simulants) were developed. Simulants are generated with similar compositions with the aim to replicate lunar soils using material science gleaned from past lunar missions. These simulants could be used to validate experimental trials for future lunar missions at various landing sites. However, these simulants are limited by our understanding of lunar soils [17] and manufacturing constraints [18].

2.1 Characteristics of regolith

Lunar regolith is a loose, clastic material coating the entirety of the Moon. Due to its environment and geology, it has properties unlike any natural material on Earth. Lunar soils are excellent heat insulators under vacuum with heat conductivity of 0.001 W/mK [23]. Thermal characteristics of lunar soil have been modelled to a limited extent [15, 24]. Lunar soil features a bulk density of 1.5–1.66 g/cm^3 [25]. Particle grains are rough (0.5 m^2/g [26]) and feature both closed and re-entrant pores. Partially due to particle shape and size distribution, lunar soil tends to easily form powder-body voids and has poor flowability [12]. Carrier et al. [13] outlined standard lunar geotechnical parameters.

The bulk composition of lunar soils typically consists of > 40% silicon, with portions of titanium, aluminium and iron. Figure 3 plots ferrite (FeO) and titanium oxide (TiO_2) against alumina (Al_2O_3) for multiple simulants, with a trend superimposed onto each chart. Alumina is a marker for the origin of lunar soils; high alumina indicates highland origin also denoted by its brighter regions and low alumina indicates mare origin displaying darker regions on the Moon. Ferrous oxide in mare soils is high (15–20%) but low in content in highland soil (~ 5%) [27]. The phenomenon leading to different lunar soil compositions is a result of exposure to space weathering processes which comprised of impact of micrometeorites, the solar wind and galactic cosmic rays. Impacts from micrometeorites create nanophase iron which reduces the available FeO in the lunar soil. Increased exposure to space weathering reduces the reflectance of the lunar soil consequently increases the alumina concentration [28, 29]. Table 2 presents mineral proportions of three lunar samples against several simulants. 14,163 is representative of endmember KREEP breccas and basalt, 67,511 is representative of endmember feldspathic crust, and 15,271 is a midpoint between the two [30]. The 'other' category includes minor chemical and mineral fractions, such as K-minerals or other silicates.

The main mineral phases in these soils are agglutinate and volcanic glasses, with mafic (silicates like pyroxene and olivine) and feldspathic (plagioclase) rocky minerals [14]. The chemical composition of lunar soils hints at their mineralogical state. Mafics are rich in iron, silicon

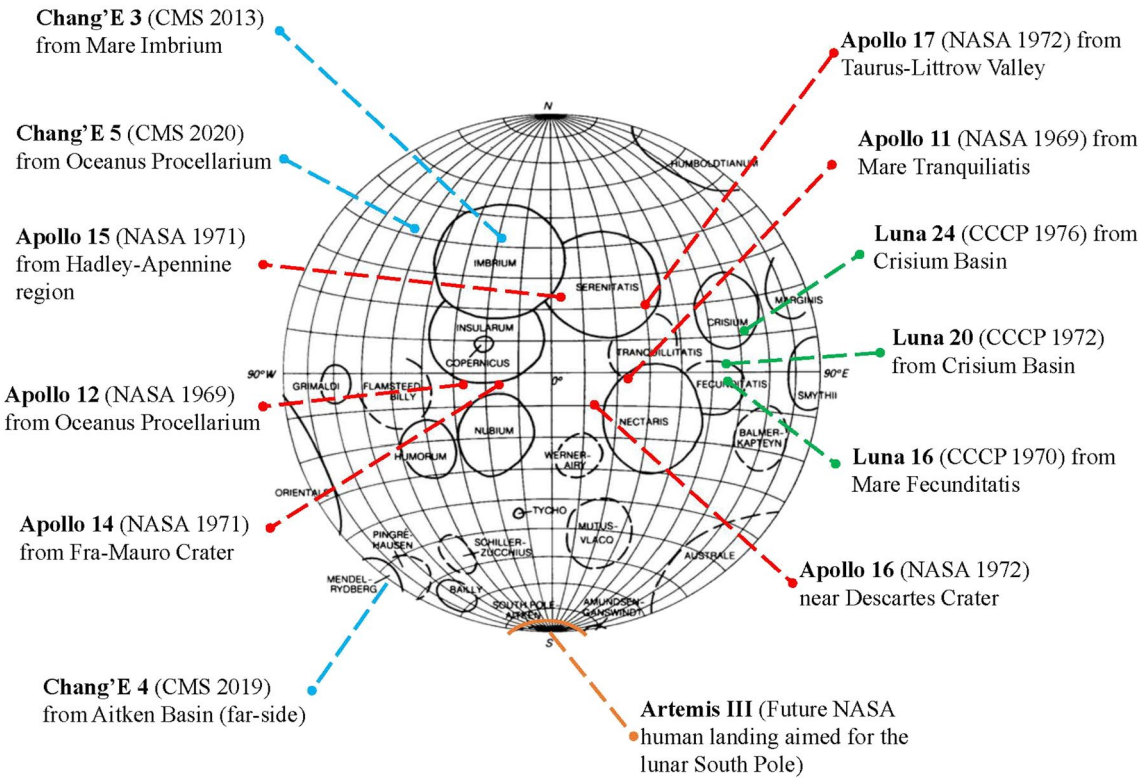

Fig. 2 Sites of Apollo [2], Luna [2], recent Chang'E landings [19–21] and the future Artemis III mission [22]. The past missions selected here returned valuable material science. Not shown are several autonomous missions from several countries planned for the Moon. The background lunar near-side map is adapted from Heiken et al. [2]

Table 1 Overview of previous soft-landing missions achieved on the Moon that underwent material science study, including a future Artemis human landing mission, commenting on the surface materials available in each site (that were sampled) [8, 33–35]

Mission	Year	Landing site	Site information
Apollo 11	1969	Mare Tranquilliatits (0.7 N, 24.3 E)	Predominantly mare composition with some feldspathic component
Apollo 12	1969	Oceanus Procellarium (3.2 S, 23.4 W)	Predominantly mare composition with some feldspathic and KREEP component
Luna 16	1970	North Mare Fecunditatis (0.7 S, 56.3 E)	Small Al, Ti–rich mare sample
Apollo 14	1971	Fra Mauro Crater (3.7 S, 17.5 W)	High-KREEP basalt with feldspathic and mare components
Apollo 15	1971	Hadley-Apennine region (26.1 N, 3.7 E)	Complex site consisting of mafic soil to Al-rich KREEP basalt
Luna 20	1972	Crisium Basin (3.5 N, 56.5 E)	Small highlands-composition sample
Apollo 16	1972	Descartes Crater (9 S, 15.5 E)	Near-typical highlands composition, with some cross-mixing from KREEP basalt regions
Apollo 17	1972	Taurus-Littrow Valley (20.2 N, 30.8 E)	Complex site; valley floor of high-Ti mare with surrounding feldspathic hills
Luna 24	1976	South Mare Crisium (12.7 N, 62.2 E)	Very low-Ti mare core sample, high in Fe and Al
Chang'E 3	2013	Mare Imbrium (19.51 W, 44.12 N)	High Ti, Fe, Al mare
Chang'E 4	2019	Aitken Basin (45 S, 180 W)	Deep SPA region; predominantly feldspathic soil with SPA pyroxene
Chang'E 5	2020	North Oceanus Procellarium (~43 N, ~50 W)	Moderate-Ti mare basalts, with some highlands and impact areas
Artemis 3	N.R	South Pole-Aitken Basin (> 175 S)	This complex region lays in the SPA highlands rim, bordering its non-mare mafic-rich interior

N.R. Not reported

and magnesium, while lunar feldspaths are rich in silicon, calcium and aluminium. Mare soils have a higher portion of mafic minerals while highland soil is dominated by feldspathic plagioclase. Due to their relative melting point

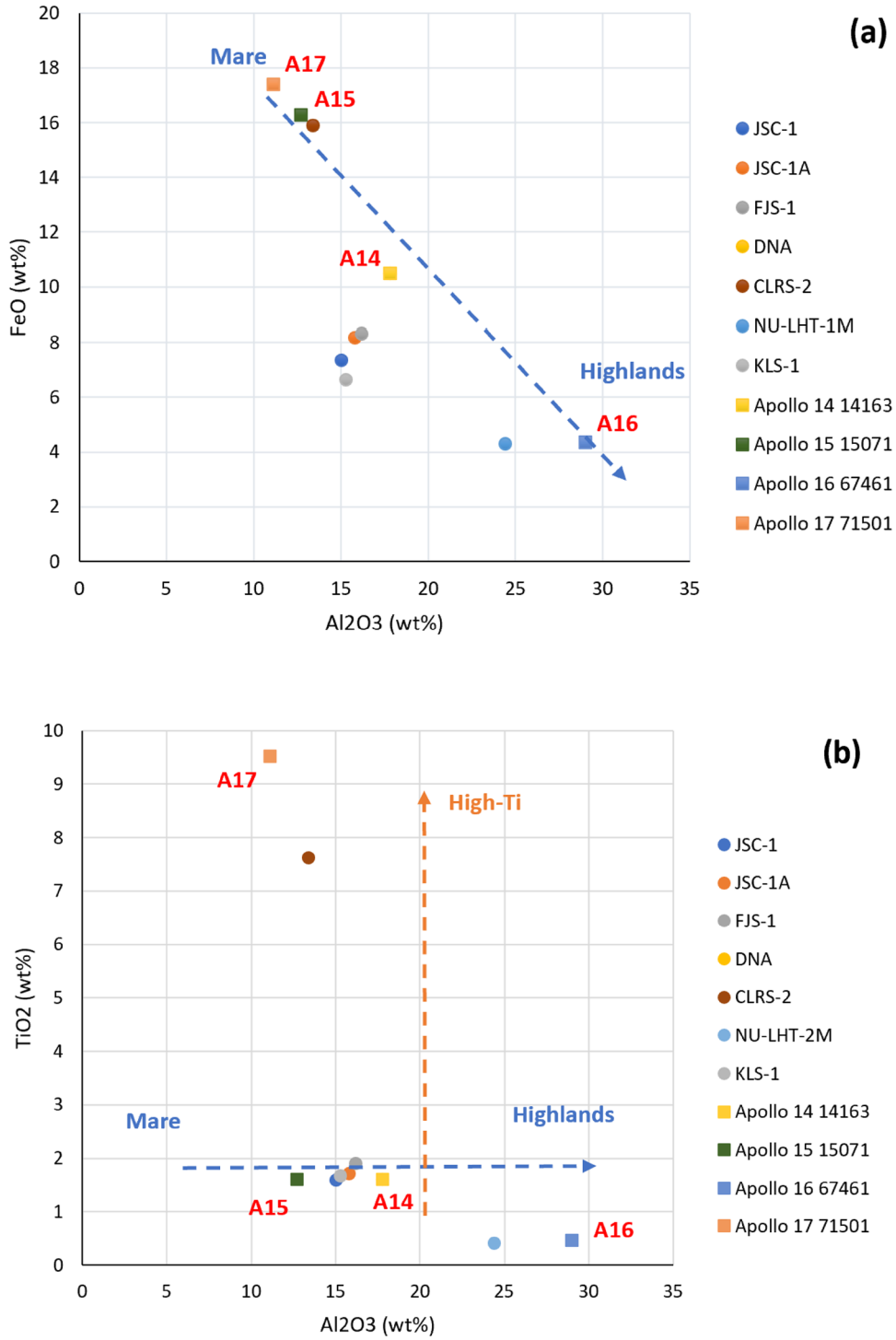

Fig. 3 Distribution of FeO (**a**) and TiO_2 (**b**) with respect to Al_2O_3 in a range of lunar simulants and soil samples. Data collected from a range of sources [9, 39, 40]

to other phases, the glasses and silicates have the most prominent influence on sintering behaviour [31]. The silicates and iron-based materials can also present high-temperature off-gassing or crystallisation reactions [32].

2.2 Simulant development

Since the lunar missions in the early 70 s, dozens of simulants have been developed to replicate lunar soil samples.

Table 2 Mineral composition (wt%) of lunar mare (top) and highlands (bottom) soils and simulants

Mineral/geologic phase	Apollo 14,163 (mare sample)	Apollo 15,071 (mare-highlands sample)	LMS-1	JSC-1	JSC-1A	FJS-1
Plagioclase	18.9	18.1	19.8	32.47	37.83	48.78
Pyroxene	16.2	22.1	32.8	15.32	19.43	25.76
Olivine	2.4	3.9	11.1	18.29	12.44	4.94
Glass	56.4	51.6	32.0	30.86	26.67	7.15
Ilmenite	0.8	1.9	4.3	0.07	0.11	3.65
Other	5.3	2.4	0	3.01	3.5	9.72
Sum	100	100	100	100.02	99.98	100
Ref	[16]	[14]	[51]	[41]	[41]	[41]

Mineral/geologic phase	Apollo 15,071 (mare-highlands sample)	Apollo 67,481 (highlands sample)	LHS-1	NU-LHT-1 M	OB-1
Plagioclase	18.1	61.2	74.4	51.87	44.35
Pyroxene	22.1	6.6	0.3	15.71	3.14
Olivine	3.9	4.0	0.2	5.79	6.27
Glass	51.6	27.6	24.7	24.07	43.22
Ilmenite	1.9	0.1	0.4	0.53	0.00
Other	2.4	0.5	0	2.05	3.03
Sum	100	100	100	100.02	100.01
Ref	[14]	[16]	[50]	[41]	[41]

Glass represents volcanic, agglutinate or natural basalt glass phases

Generally, the particle sizes of most simulants range within 0.002–2 mm, as received, which are similar to lunar soils or can be milled to match. The similarity in the chemical composition is a more complex topic.

JSC and NU-LHT were among the first mare and highland simulants developed respectively. JSC-1 was replicated against low-Ti mare sample 14,163 using basalt glass. NU-LHT was designed based on the average of Apollo 16 highland samples utilising glass produced from plasma melting to simulate agglutinate particles [36–38]. Both were designed to replicate the chemical and geotechnical characteristics of Apollo soil samples but were produced before detailed mineral compositional data was available [14]. The discrepancy between mineral content in these soil samples can be seen in Table 2.

More mare simulants were produced based on Apollo samples and JSC-type simulants [18]. FJS-1, KLS-1, CLRS (or CAS) and DNA were mare simulants derived from volcanic ash or basalt from terrestrial sources [41], and featured similar chemical composition to JSC-type simulants [42–45]. All (aside from CLRS-2) contain low portions of titanium. Glass in these simulants is represented by glass-rich basalt. Their specific compositions are outlined in other works [31, 46–48].

Several recent simulants have been generated with a tuneable chemical, mineral or grain composition that can accommodate site-specific mixtures. LHS-1, LMS-1, TUBS-T, TUBS-M were root examples of highland and mare simulants [49–51]. LHS-1 was based on highland soil 67,641 while LMS-1 was derived from 24,999 [52]. Higher-fidelity features, such as synthetic agglutinates, iron particles and ice-bearing soil are in development [49, 53, 54]. Toklu et al. presented the most comprehensive list of lunar simulants as of time of writing [9].

While early simulants (e.g., JSC, NU-LHT) were originally designed for studies into granular mechanics, they were also utilised for many engineering needs. More recent simulants have been developed with a wider range of soil characteristics, including mineralogical and morphological features. However, simulants are unable to replicate every aspect of lunar soils as data on lunar soils is not comprehensive. Thus, further research on chemical, thermal and physical characteristics of lunar materials are required to improve the performance and fidelity of simulants [9]. At this stage, there are no particular simulants that are key in material engineering particularly for any particular AM technique to the knowledge of the author which was supported by the results of the mechanical properties of different regolith simulants in Table 6.

3 Bulk sintering

3.1 Overview

Production of construction materials for structures, space shielding and landing pads are a facet of industrial activities

necessary for a lunar outpost. Lunar regolith is a readily available resource that can be used as feedstock. Where Earth-bound concrete uses liquid water in cementitious materials, liquid binders may sublimate in the lunar environment. There are very few studies on Moon-bound concrete utilising sulfur concrete and polymer concrete which may not require water as the solution which proves its viability. However, certain materials involved not available in-situ may raise the transportation cost significantly [55]. Sintering of lunar soil can be employed to minimise liquid binders.

The choice of energy source or sintering method determine the sintering profile applied to lunar soil and the final material characteristics. Lunar soil is a multiphase material, featuring minerals or glasses of varying melting temperature. Liquid-phase sintering of lunar soil or simulant is more feasible than solid-state sintering [46]. Bulk sintering methods vary mainly in their energy source and aim to consolidate a relatively large volume of material and will be discussed here. Radiant sintering utilises infrared radiant heat, microwave sintering uses electromagnetic radiation, and spark plasma sintering utilises high-power electrical currents. Other techniques discussed in a later section employ focused energy and additive manufacturing strategies to consolidate material.

Sintering trials commonly focus on a narrow range of soil characteristics. The porosity of a solid material determines its mechanical properties, particularly its strength and failure mode. Particle size distribution influences the bulk density of a powder body, which inherently affects the sintered density. Although sintering increases bulk density by rearranging grains, increasing packing density from the outset provides higher efficiency. Mineral and chemical compositions affect the proportion and components of the liquid melt influencing the sintering process.

Lunar soil is expected to melt between 1100 and 1380 °C [24]. A study examining the heat flow of a small number of Apollo samples displayed no thermal reactions below 1000 °C [15]. The heat flow characteristics for a range of simulants were previously reported [40, 56]. In a Hot-Stage Microscopy trial conducted at 1200 °C with a heating rate of 10 °C/min, JSC delaminated from a pre-sintered substrate 44 °C below the sintering point of the simulant [57]. This is due to surface tension forces surpassing the weight of the substrate resulting in layer delamination. In contrast, JSC smoothly adhered to a chemically inert substrate [57]. Bloating effects were observed above 1150 °C, worsened by slower heating rates [58].

3.2 Radiant heat sintering

Radiant sintering (or ambient sintering) involves using heating elements to irradiate and heat a large sample volume. As radiant sintering is the most straightforward, this section will act as a baseline for newer or upcoming techniques.

Lunar soil sintering focuses on generating a liquid phase from lower melting temperature constituents, forming bridges between particles of higher melting temperature and enabling diffusion or melting in these particles. In lunar soils, this inter-grain matrix was formed from glasses and mafic minerals binding plagioclase grains [31]. Hintze et al. published a comparison using SEM characterisation on radiant sintering across several simulants (FJS-1, JSC-1A, NU-LHT-2 M, OB-1) [31]. Highland simulants had higher sintering temperature and lower densities than mare simulants under similar conditions. The portion of available glass in simulants was the most significant factor in this study [31].

3.2.1 Discussion of process 'window'

A study into the sintering of small FJS-1 samples revealed that sintering temperature had the greatest impact on final sample strength. 25 °C below the melting point was found to be ideal for sintering [59] (~ 1100 °C for FJS-1 [56]). A longer heat time was required to enable even heating across larger samples [59]. For highland simulants, sintering temperature has exceeded 1300 °C [60]. In several studies, employing a vacuum atmosphere reduced the sintering temperature by 25–100 °C [32, 46, 59].

Table 3 outlines the sintering temperatures for a range of trials. Hot Stage Microscopy can quantitatively define this temperature as the 'half-ball' point, a stage when a semi-circular section is formed on the sample [57]. Other studies performed parameter sweeps and took microscopy observations on the resulting sample structure or measured the bulk density of the sample. Despite sintering temperatures ranging between 1100 and 1200 °C for mare simulants, modifying heating rate could alter this temperature range. The working temperature range defined in Table 3 is described as liquid-phase sintering and partial melting [61] which are the dominant mechanisms behind consolidation of the simulant.

3.2.2 Mechanical properties of sintered samples

Radiant sintered samples achieved compressive strengths in the range of 30–200 MPa and porosity below 1% (Table 4). The highest compressive strengths were achieved with high pre-pressing forces (100–200 MPa), sieving of particle sizes (< 200 µm) or inserting a fine ilmenite powder (< 50 µm). However, one trial achieved 218 MPa average compressive strength using only moderate (4 MPa) compression force and no additives [62].

Finer simulant batches proved to facilitate higher sintered densities. The finer samples had significantly higher compressive strength (~ 230 MPa) and density (> 98%) compared to coarser samples (~ 100 MPa, ~ 90%) [62, 63]. Improved

Table 3 Sintering temperature range from a range of radiant sintering trials

Simulant	Working temperature (°C)	Method	Notes	Refs
JSC-2	1185–1250	Hot stage microscopy	10 °C/min on inert substrate	[57]
	1200–1250		10 °C/min on pre-sintered substrate	
JSC-2A	1195–1280	Hot stage microscopy	10 °C/min on inert substrate	[58]
	1220–1300		50 °C/min on inert substrate	
FJS-1	1150–1200	Sintering + electron microscopy (EDS)	Inert substrate	[31]
JSC-1A	1100–1200			
NU-LHT-2 M	> 1150			
OB-1	> 1150			
CLRS-1	1050	Sintering + electron microscopy (EDS)	Inert substrate in vacuum	[32]
	1100–1150		Inert substrate in air	
CLRS-2	1100	Sintering + electron microscopy (EDS)	Inert substrate in vacuum	[48]
TUBS-M	1150–1200	Sintering + bulk density data		[60]
TUBS-T	1350–1500			
HIT-L-1	1219–1300	Hot stage microscopy	5 °C/min on inert substrate	[61]

Table 4 Mechanical properties from a range of radiant sintering trials

Simulant	Sintering temperature (°C)	Atmosphere	Compressive strength (MPa)	Elastic modulus (GPa)	Hardness (HV)	Sample condition	Notes	Refs
JSC-1A < 212 µm	1200	Nitrogen	232	10.9	1027	0.41% porosity		[63]
JSC-1A > 212 µm			103.2	5.98	1027	8.44% porosity		
FJS-1	−25 below melting point	Air	33.3–37.8	6.1–13.8	–	1.6–1.8 g/cm^3 density	Brick-sized sample	[59]
JSC-1A + fine mix	1125	Air	150	27.5	–	–	255 MPa pre-press	[46]
JSC-1A + coarse ilmenite mix			50–100	10–22.5		–		
JSC-1A			150	25		22.7% porosity		
	1100	Vacuum	100	17.5		12.8% porosity		
JSC-1A < 212 µm	1120	Nitrogen	218.8	8.4	–	2.6 g/cm^3 density	–	[62]
JSC-1A > 212 µm			84.6	5.5		2.2 g/cm^3 density		
HUST	1050	Vacuum	37.48–42.34	–	–	82.54% dense	25 MPa pre-press	[64]
			31.82–38.47			78.89% dense	No compaction	
			44.75–48.12			84.86% dense	10 MPa pre-press and 25 kPa hot-press	
			58.45–67.68			89.83% dense	25 MPa pre-press and 25 kPa hot-press	

mechanical properties in one study were attributed to finer initial packing due to the inclusion of fine powder into a simulant [46]. Increasing ilmenite content did not correlate to higher mechanical strength in sintered samples [46, 48].

With increment of sintering temperatures, the strength of the final product increases [59]. Hot pressing is an alternative method of compacting powders, pre-heating to improve the density of sintered samples influenced by the rearrangement of grains [64]. Crystallisation reactions in iron-based minerals have been observed to degrade the mechanical properties of sintered samples under high (> 1200 °C) temperatures [48]. Improving mechanical properties is essential for the structural needs to withstand constant bombardments of micrometeorites, radiation shielding and static loading conditions. Some examples of

structures the require high mechanical properties are road surfaces, dust mitigation and launch or landing pads [62].

3.2.3 Vacuum effects

Several studies have commented on the effect of lowering sintering temperatures required by conducting experiments under a vacuum environment [32, 48]. Oxygen environments can react with iron-based minerals present in lunar soils, weakening their cohesion [46]. Additionally, in silicate-based minerals under high temperature, off-gassing can occur. Song et al. [32] observed that this gas release bloated the sample and resulted in a porous material. The material quality is highly sensitive and requires appropriate control during the vacuum sintering process.

3.3 Microwave sintering

Microwave sintering is a method that uses electromagnetic radiation in the 300 MHz-300 GHz band to heat a powder sample through dielectric absorption. The potential energy efficiency of microwave heating and the ability to tune the microwave frequency to couple with specific phases in lunar soil are advantageous to construction materials fabrication [65]. Key parameters to this process are microwave field frequency and strength, penetration depth and power density of this process [26]. Microwave heating is also sensitive to material properties, compaction and their high-temperature behaviour.

3.3.1 Mechanism for microwave absorption into lunar soil

Lunar soil couples extremely well with microwave radiation [26]. From initial studies, it is believed that the defects and impurities caused by weathering or solar volatiles contribute to the increased microwave absorption by lunar soil [65]. A more recent conjecture is that nanophase iron particles, rather than defects, act as effective microwave absorbers [26]. However, experimental trials with Apollo samples did not support this theory as experimental trials are conducted by Barmatz and team [66] on the permittivity across numerous regolith. The results obtained show a lack of correlation between the presence of nanophase iron particles and microwave permittivity. A dramatic increase in the heating rate of lunar simulants was observed around 400 °C, suggesting a phase change in the interior part of the sample that led to more effective absorption [67]. Increased ilmenite content (Table 2) weakly improved soil microwave absorption [68]. Mare soils are observed to have higher dielectric loss tangents than highland soils which is due to higher wt% of FeO and TiO_2 in the mare samples as seen in Table 2 [66]. Nearly all microwave sintering studies have focused on the 2.45 GHz spectrum. This frequency happens to be effective at heating lunar soils or simulants but is not specifically optimised for this purpose. Sintering was found to be feasible at microwave input powers of 0.8–1 kW [69].

3.3.2 Microwave heating 'runaway'

Microwave heating if uncontrolled can result in overheated and non-uniform samples. As a material heats up, its absorbency to microwave radiation can increase dramatically alongside the power density of the heating process. This results in microwave radiation heating an increasingly small volume of material at an accelerating rate [26]. However, using an additional radiant heat source will provide extra stability in the process. Susceptors were found to reduce the total energy required to melt JSC-1 in comparison to microwave heating on its own, particularly in the preliminary heating process [70]. Susceptors bridged the poor room-temperature heating of microwave energy, and in raising the ambient temperature of the furnace enabled more uniform heating and sintering [70].

3.3.3 Computer simulation of heating process

Modelling and experimentation were done with a large volume (8.5 kg) of JSC-1. Hybrid microwave and susceptor heating resulted in a wide, shallow heating region of 1 cm deep in the powder body [71]. This depth is rationalised as a sintering penetration depth for 2.45 GHz radiation, at least for mare simulants. A more recent simulation study revealed a relationship between microwave absorption and the sample dimension [72]. This simulation model was applied to the analysis of a low-power (250 W) experimental payload design [73]. The correlation of cavity design, microwave frequency and the sample dimension should be studied to optimise microwave systems. Further simulation should integrate high-temperature behaviour of regolith to better understand how these factors can influence the heating process.

3.3.4 Mechanical properties of microwave-sintered samples

Table 5 outlines the mechanical properties of microwave sintered specimens. Most microwave sintered simulants achieved in the range of 10–50 MPa and porosity as low as 9%. One trial achieved compressive strengths as high as 70 MPa, through employing high (1300 °C) peak temperatures with a small portion of ilmenite additive [74]. Increasing ilmenite content to a significant (> 30%) proportion of the sample degraded the strength of the product [74].

Compression testing of susceptor-assisted microwave samples revealed two common failure modes; longitudinal splitting failure with a strength of 30–45 MPa and shear failure with a strength of 20–25 MPa [75]. Shear failure

Table 5 A series of microwave sintering trials and their resulting mechanical properties

Simulant	Temperature (°C)	Atmosphere	Compressive strength (MPa)	Elastic modulus (Gpa)	Hardness (Gpa)	Sample condition	Notes	Refs
KLS-1	1080	Air	12.6	–	–	35.13% porosity	Hybrid heating	[40]
	1100		20.8			32.15% porosity		
	1120		37.0			27.84% porosity		
CLRS-1 + 4.6% ilmenite	1050	Nitrogen	30	–	–	Low porosity	–	[74]
	1150		45			Low porosity	–	
	1300		74			Low porosity	–	
CLRS-1 + 28.5% ilmenite	1050		70			Pores/cracks	High mechanical variation	
	1150		30			Pores/cracks		
	1300		55			Pores/cracks	–	
JSC-1A	–	Air	–	64.8	6.78	2.81 g/cm^3 density	0.6 kW fixed input power	[69]
				76.3	7.75	2.98 g/cm^3 density	0.8 kW fixed input power	
				70.0	7.73	3.02 g/cm^3 density	1 kW fixed input power	
FJS-1	1125	Air	20–45	–	101.39–106.23	8.63–9.64% porosity	Hybrid heating	[75]
	1100				89.84–97.80	9.05–9.92% porosity		
	1075				89.13–94.31	10.06–11.51% porosity		

occurred in a radiant sintering study because of lower porosity in the sample walls [63]. Mechanical properties of microwave-sintered samples are lower than in the case of radiant heat sintering (Table 4) due to factors such as difficulty in controlling heating and cooling rates for microwave than for radiant heating. This major factor will lead to uneven heating and poor control of temperature rate cause thermal cracking in the samples contributing to lower compressive strength compared to radiant sintered products [76].

The potential to use microwave heating as a thermal post-treatment for radiant sintered samples has been examined but results were inconclusive and require closer examination [77]. Further research is required towards depth penetration and uniformity of the sample.

3.4 Spark plasma sintering

Spark plasma sintering (SPS) utilises a combination of pulsed electric current, external pressure and rapid heating rate in a vacuum to sinter powders (Fig. 4) [78]. The current flows through the sample producing an electric field allowing heating to occur through the sample volume. Utilisation of both electric field and applied pressure permits high heating and cooling rates (at 1000 °C/min) [79] and shorter sintering periods [80, 81]. SPS was also observed to produce a higher density of sintered samples compared to radiant sintering [82, 83]. The external pressure promotes the melting of grain surfaces and densification through plastic deformation [84]. Similarly, SPS requires lower sintering temperatures compared to radiant sintering processes (Table 4).

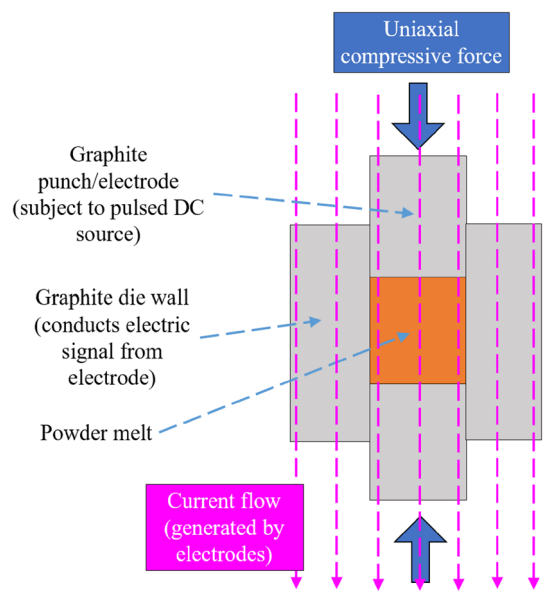

Fig. 4 Diagram of spark plasma sintering plant, adapted from Igorevich et al. (Credit: [85])

Table 6 Summary of process parameters and mechanical properties of SPS specimens

Simulant	Average particle sizes (μm)	Temperature (°C)	Pressure (MPa)	Heating rate (°C/min)	Dwelling time (min)	Atmosphere	Density (g/cm^3)	Hardness (GPa)	Porosity (%)	Refs
FJS-1	100	800	100	100	10	Vacuum	2.19	–	–	[87]
		900					2.55	–		
		1050					2.9	8.51		
JSC-1A	1.67	550	30	100	–	Vacuum	2.755	5.49	–	[91]
		700	50				2.813	6.01		
LHS-1	50	1025	80	100	20	Vacuum	2.532	4.345	23.7	[92]
	50	1050					2.616	7.11	21	
	50	1075					2.704	7.287	11.7	
LMS-1	50	1050					2.817	7.179	21.6	
EAC-1A	22	1050					3.04	7.081	4.3	
	50	1050					2.831	7.375	15.5	
	100	1050					2.795	6.443	21.4	
FJS-1	70	900	50	100	10	Vacuum	2.5–3.0	–	–	[93]
		1000	50		10					
		1000	100		10					
		1030	50		10					
		1030	50		20					
FJS-1	100	1000	100	200	20	Vacuum	2.75	–	7.57	[88]
		1050	50	50	5		2.82		5.18	
		1050	75	100	20		2.93		2.21	
		1050	100	150	30		2.93		2.07	
JSC-1A	28.55	900	15	100	3	Vacuum	2.84	–	5.87	[94]

3.4.1 Influence of process parameters on mechanical properties

Process parameters heavily influence the mechanical properties of ceramic samples [86]. Table 6 demonstrates the influence of SPS process parameters on sintered samples. Zhang et al. [87, 88] studied the effects of this technique on the densification, microstructural evolution and mechanical properties by varying process parameters. Contour maps displaying density and dense area percentage with respect to temperature and pressure are shown in Fig. 5. Results show that density as high as 2.9 g/cm^3 can be achieved at 1050 °C and 2.55 g/cm^3 density at 900 °C under a uniaxial pressure of 100 MPa. Density of normal concrete and clay brick are 2.4 g/cm^3 and between 1.8 and 2.0 g/cm^3 respectively; demonstrating higher density observed in SPS samples in comparison to other conventional materials [89, 90]. Given sufficient temperature and pressure, higher dwelling time and lower heating rate were observed to increase the dense area percentage of the sample.

3.4.2 SPS at low temperatures

SPS samples in a vacuum were observed to densify at as low as 550 °C and 700 °C with pressure of 30 MPa and 50 MPa respectively [91]. At these parameters, 98% density was obtained at 700 °C under a pressure of 50 MPa. Although the hardness of the sample was recorded at 6.01 GPa, which is a slightly lower value than measured in other studies [86, 92] due to lower temperature and pressure, the author mentioned that the properties would be suitable for structural materials in lunar habitations.

3.4.3 Influence of particle size

Poporich et al. [92] conducted experiments detailing the effects of various particle sizes on densification, porosity and hardness. The finest sample (< 22 µm) resulted in the highest density (3.04 g/cm^3) and lowest porosity percentage (4.3%) among the samples tested. Smaller particle sizes contribute to higher surface energy which leads to higher densification through the SPS technique. The tensile stress and strength of the sintering necks increase with finer particle sizes. The < 50 µm sample batch features high hardness, corresponding to a wider range of particles. Hardness does not increase with powder milling but powders are more homogenously distributed.

3.4.4 Influence of powder composition

A study by Poporich et al. [92] covered the effects of different simulants (EAC-1A, LHS-1, LMS-1) utilising SPS with 1050 °C, 80 MPa pressure and a maximum particle size of 50 µm. All three simulants display similar densification with the LHS-1 sample falling short of the other two simulants. The average porosity is lower for EAC-1A and LMS-1, but LHS-1 possesses higher heterogenous porosity along with a highly densified part. All simulants were composed of very similar amount of oxides and minerals, leading to a similar sample hardness. LMS-1 has more iron titanium oxide phases, leading to a larger hardness variation through the sample.

4 Focused sintering

This section provides an overview on sintering methods that adopt a focus, directed energy source to fuse simulants. Specifically, pertinent process parameters for selective laser sintering and solar sintering are examined for different types of lunar simulants. While these methods use the same fundamental sintering process, they lead to unique sintering profiles and therefore material characteristics. These methods are further assessed on their viability as an additive manufacturing process.

4.1 Laser sintering

Selective laser sintering or laser fabrication involves rapidly heating a small volume of powder using a high-power laser

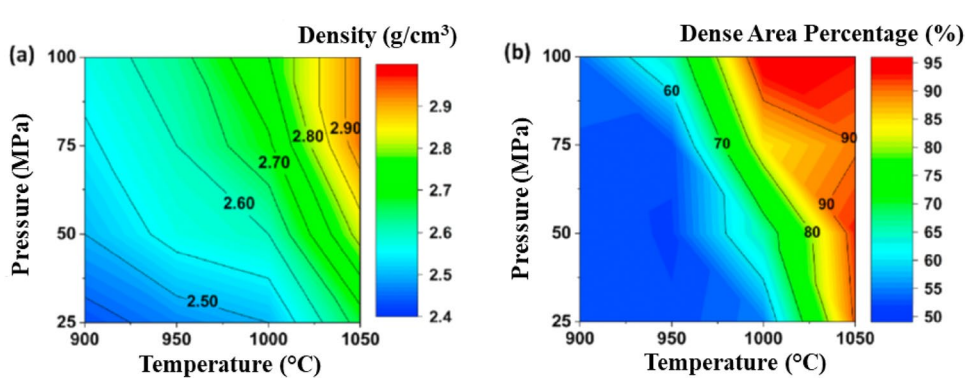

Fig. 5 Contour maps of **a** density and **b** dense area percentage as a function of sintering temperature and pressure (Credit: [88])

to fuse as solid material. Extreme cooling rates (> 1000 °C/s) are a characteristic of this method. Laser fabrication trials discovered that lunar soil was an excellent absorber of laser energy in the 1000 nm range. Mare simulants had from 75% (CLRS-1 [95]) to ~ 92% (JSC-1A [96]) absorbance around 1000 nm. Laser power as low as 50 W was able to fully melt JSC-1A [97].

4.1.1 Influence of particle characteristics on the sintering process

Geotechnical properties of simulants have a significant effect on laser fabrication. Laser sintering of metals strive to use powder of very narrow particle size ranges (10–50 µm) of high sphericity to achieve minimal porosity and high mechanical performance. Higher porosity in powder bodies can disperse laser irradiation and entrap gas pockets, inhibiting densification [98]. The poor packing density of simulants made laser sintering difficult. A study using a 75 µm spot size to fuse < 1 mm particles noted a final product with high porosity [99]. Other trials have used pre-pressing [95] or particle sieving [100] to minimise this.

4.1.2 Formation of melt tracks

Solid, evenly sintered tracks produced by laser infiltrating powder material are desirable for strong materials. The key variable to achieve this is scan speed, given laser parameters and powder characteristics. In lunar sintering trials, ideal scanning speeds ranged from 25 to 250 mm/s depending on other variables. The actual fusion zone can extend beyond the laser spot diameter laterally and into the powder bed. The fusion zone width, rather than spot diameter, assisted in determining hatch spacing and layer thickness in AM processes. With the right conditions, this fusion zone penetrated up to 4 mm into the simulant powder body [101].

A study by Goulas et al. [100] observed solid particles protruding from the melt pool generated by a laser track. These were identified as large particles that were not sufficiently melted, as well as olivine and plagioclase crystals with a high melting temperature [100].

4.1.3 Discussion of process 'window' for laser sintering

Figure 6 presents a range of key trials with respect to their reported ideal laser power and scan speed. While these are good indicators for laser fabrication, there are other factors that can alter the process window. Table 7 showcases several key parameters used in laser sintering trials and the corresponding results. While process windows in these trials are within similar ranges, the choice of simulant and other parameters affected this exact window. Alumina substrate more readily allowed simulant adhesion in terms of laser power and scan speed than stainless steel [102]. This is due to the high thermal conductivity of stainless steels and the low wettability with simulants. Other studies reported that the laser penetrated into the steel generating a localised melt plume that ejected the chemically incompatible simulant [103].

4.2 Solar sintering

Most techniques of solidifying lunar regolith require large, external energy supplies and, thus, would require the transportation of power plants or solar panels to the moon. An alternative is to directly utilise the Sun, an unlimited existing source of energy. The solar sintering process involves focusing sunlight to heat up sand or regolith to craft building elements to construct structures as future habitats on the Moon

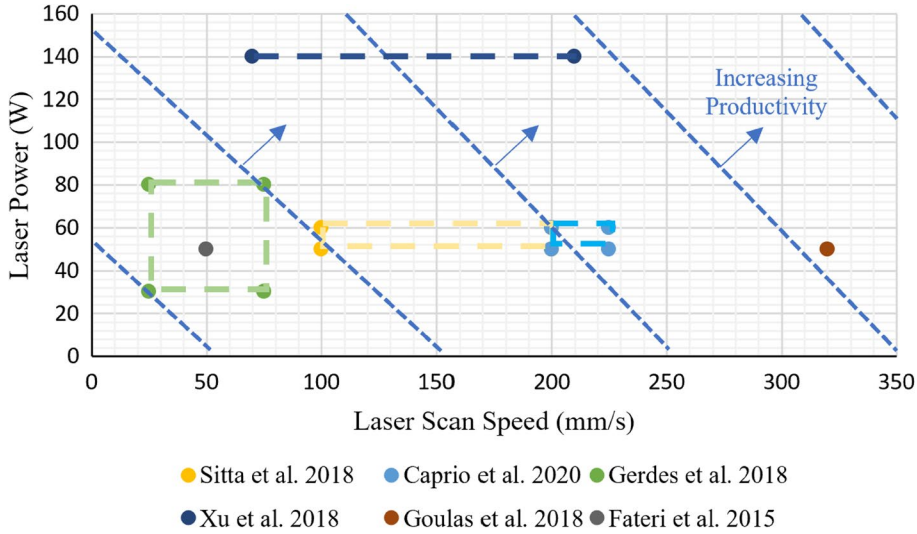

Fig. 6 Process window for a range of regolith laser fabrication studies; the dotted rectangular regions represent windows in which a solid material was 'successfully' fabricated [95, 99, 100, 102–104]

Table 7 Summary of process parameters used in a range of laser fabrication studies

Simulant type	Simulant size fraction	Power (W)	Laser spot (µm)	Scan speed (mm/s)	Hatch spacing (µm)	Layer thickness (µm)	Energy flux (J/mm^2)	Substrate	Refs
JSC-1A	< 63 µm	50	15	50	200	100–300	5	N.R	[104]
NU-LHT-2 M	As-is	50–60	50–300	100–200	75	150	2.96–4	Refractory clay	[99]
JSC-1	As-is	30–80	70	25–75	50	150	8–64	Alumina	[102]
		70–100		25–50			28–80	Stainless steel	
CLRS-2	As-is	140	121	70–420	N.R	1910–4120	0.6–0.9	Alumina	[95]
JSC-1A	< 125 µm	50	80	200–400	170–250	150	0.5–5	Soda lime glass	[100]
NU-LHT-2 M	< 250 µm	50–60	60	200–225	75	150	3–4	Refractory clay	[103]

Each study utilised a laser of ~ 1070 nm in a pressurised inert atmosphere (Argon)

N.R. Not reported

[105, 106]. This method is desirable for lunar construction, as it effectively takes advantage of local resources.

4.2.1 Solar concentrators and process 'window' of solar sintering

Physical Sciences Inc. (PSI) [107] suggested the use of a solar concentrator array to direct sunlight to sinter regolith directly. Nakamura and Smith [107] utilised a solar concentrator array connected to a quartz rod (36 mm diameter) to transfer solar radiation onto the regolith inside the reactor at 33.3% efficiency. Sintering onto a Tephra substrate occurred in a temperature range of 1000–1100 °C. Exceeding 1150 °C will form molten slag instead. Further trials were conducted to successfully obtain a sintered pad (15 × 15 in) with a solar heat flux of 70 W/m^2 and rastering speed between 1–2.35 mm/s as shown in Fig. 7. However, mechanical properties were not reported in the work.

Despite the efficiency shown in the experimental trial, dust effects affected the inlet optics which reduced the performance of the system by 6%. Employing dust cover to counteract the issue, however, lowered the power output of the primary concentrator by up to 10% [107].

Although the PSI team managed to conduct the experiment on the field, weather conditions experienced on the Moon are substantially different than on Earth. Solar cycle on the Moon is a key criterion to consider when utilising solar sintering and would require more trials. Another possible point of research is the time consumed to setup and sinter at specific locations as compared to sintering building elements and constructing infrastructures separately.

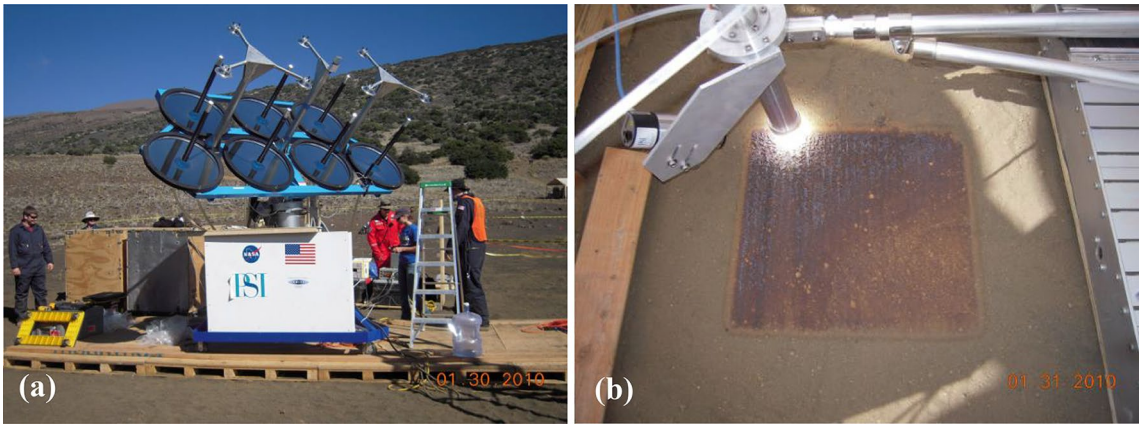

Fig. 7 Images of **a** fuly assembles solar concentrator system, **b** solar sintering on Tephra surface. (Credit: PSI [107])

5 Additive manufacturing applications

This section examines how laser, solar and microwave sintering methods could assist additive manufacturing (AM) applications. Direct Ink Writing, using a liquid binder, is introduced as a sintering method in this section. This section outlines the requirements for each manufacturing process and identifies reasonable applications in a lunar environment for each.

5.1 Applications of 3D laser sintering

Laser sintering is an additive manufacturing that utilises a laser that provides a concentrated heating beam to trace over the surface of a tightly compacted powder melting and sintering the powder. Once a layer is finished, a piston will move the platform down and a roller is used to spread excess powder on the build platform. These steps will be repeated layer-by-layer until a completed 3D part is built [108]. Laser sintering is not economically feasible as a bulk construction method relative to other sintering approaches. Due to the high geometrical accuracy and other environmental properties, laser fabrication may be viable for producing small precision-engineered mechanical parts in low-stress, high-temperature or corrosive environments [104].

5.1.1 Mechanical properties of laser-sintered samples

Laser fabrication achieved both solid, glassy samples and highly porous samples. Using a narrow 50–150 µm grain distribution and direct deposition methods, an early study achieved a 95% of the theoretical density of the simulant (JSC-1AC) [97]. In contrast, another study reliably printed samples of ~ 50% density [100], where higher density sacrificed print quality and reproducibility. The former was a brittle, hard material. The latter exhibited behaviour similar to ceramic foams and could absorb a larger amount of mechanical energy than a stiff brittle material [100]. Up to a limit, as laser energy increased, these materials became stiffer and withstood a higher peak stress [100]. Another study utilised a high feed rate for outer part contours and a lower feed rate for the inner structure [104].

According to Table 8, laser-fabricated samples achieved compressive strengths of up to 30 MPa, and with porosities as high as 65% to as low as < 5%. The studies that achieved low porosities were only conducted on very small samples [97, 102], and mechanical testing data was not available. Consequently, the highest compression strength of 30 MPa was achieved on a sample with 35% porosity [103]. This high porosity could be attributed to the large particle size distribution of simulants.

A recent study examined the flexural and compressive properties of several simulant types that underwent laser sintering [101]. Laser-based paving of large surfaces for dust mitigation purposes was found to be feasible but would be very limited in loadbearing capacity.

5.2 Applications of 3D solar sintering

Similar to laser sintering, 3D solar sintering instead utilises concentrated sunlight as the heating source to sinter the powder. A heliostat is used to focus the sunlight on a mirror-based concentrator to direct the concentrated solar light towards the powder layer-by-layer generating the 3D desired component [47].

5.2.1 RegoLight system

The RegoLight project investigated the feasibility of building lunar habitation, providing ideal shelter for humans to live on the Moon, protected from harmful radiation. Teams

Table 8 Summary of mechanical properties from samples produced in laser fabrication studies

Simulant type	Simulant size fraction	Energy flux (J/mm^2)	Compressive strength (MPa)	Elastic modulus (GPa)	Hardness (HV)	Bulk density (%)	Notes	Refs
JSC-1AC	50-150 µm	2.12	–	–	500	92–95	Direct deposition method	[97]
JSC-1A	< 63 µm	5	–	–	1245 (HVIT)	–	Smooth, geometrically accurate sample	[104]
NU-LHT-2 M	As-is	2.96–4	16.6–31.43	–	~ 669–748	–		[99]
JSC-1	As-is	8–64	–	–	–	< 96		[102]
JSC-1A	< 125 µm	0.5–5	< 4.2	< 0.287	657	44–49	Smooth, geometrically accurate sample	[96]
NU-LHT-2 M	< 250 µm	3–4	< 31.4	–	680–700	35–65		[103]

from the European Space Agency (ESA) and the German Aerospace Center (DLR) were involved in the project and outlined the possibility of the use of solar sintering to create the building blocks and various other applications for the ISRU method [106]. The DLR team clarified their printing procedure which begins with regolith powder (JSC-2A) deposited layer-by-layer using the feeder. Xenon arc lamps are used to provide power of 21 kW and flux density of 4.3 MW/m^2. As regolith simulant is deposited, a 3-axis table was utilised to manoeuvre the test bed under the concentrated sunlight following a specific design layer-by-layer. As a layer is sintered, a cooled wall was used to prevent unneeded sintering as continuous exposure may lead to overheating of the hardware. Figure 8 illustrates a schematic design of the experimental setup [47].

5.2.2 Process optimisation of solar sintering

Sintering process parameters were optimised according to visual observations on sintered and molten lunar regolith simulants. Warping of the sintered part could be mitigated by printing the first layer of the regolith simulant on a porous substrate to promote bonding and avoid warping. Optimised scanning parameters were obtained at 48 mm/s scanning speed, 14 mm spacing between sintered lines and 0.1 mm layer thickness. Meurisse et al. manufactured $20 \times 20 \times 20$ mm^3 for compression tests and results obtained show average compressing strength of 2.31 ± 0.30 MPa and Young's Modulus of 0.14 ± 0.06 GPa, based on 12 samples. Open pores (up to 0.6 mm) are observed due to poor bonding between layers and closed porosity (up to 0.45 mm) indicated that high-volume fraction material is melted during sintering [47].

A study fabricated interlocking building elements using solar sintering methods and developed a simulation of a lunar structure. Bricks of reasonable geometrical accuracy for concave/convex shapes were achieved. It was concluded that solar sintering could be used for bulk shielding of lunar structures due to its porous brittle nature [109], though, further vacuum testing is needed. Solar sintering in theory can operate at 10–100 cm^3/h and could produce construction materials in a reasonable timeframe. Solar sintering requires minimal operating power but will require a direct source of sunlight and maintenance (dust removal) of the solar arrays.

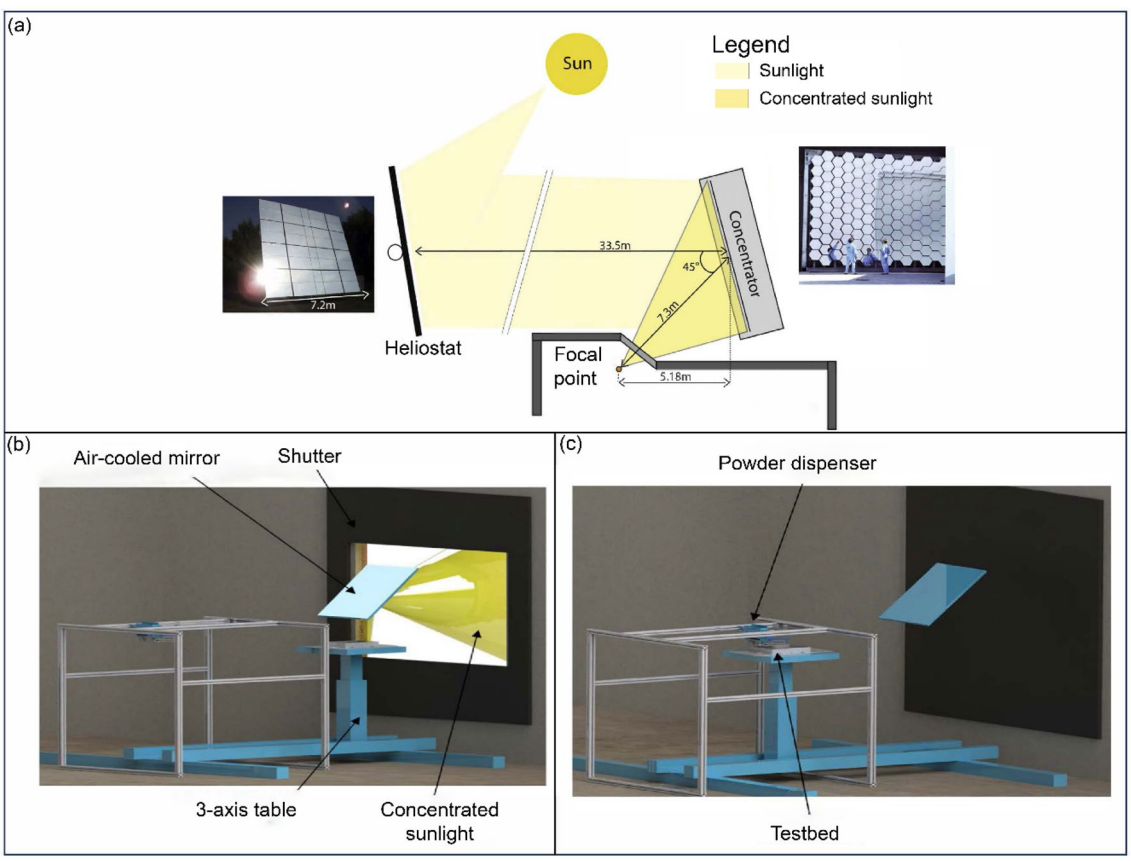

Fig. 8 **a** Schematic of the experimental setup of the RegoLight system; **b** during sintering and **c** during deposition of a new layer of regolith (Credit: DLR Team [47])

5.3 Applications of 3D microwave sintering

Microwave sintering can also be applied to additively produce 3D structures. 2.45 GHz microwave radiation has a ~ 1 cm sintering penetration into lunar simulant [71]. Microwave radiation could be used for paving surfaces for landing pads or dust mitigation [26, 110].

Figure 9 presents two strategies for additive-printing microwave cavity design. First strategy showcases a microwave printhead cavity developed for the ATHLETE which takes in lunar soil as the feedstock and heat/melt in a tube while depositing in layers creating the 3D structure [111]. Secondly, a microwave emitter/source could be utilised to microwave sinter lunar soil. These microwave emitters will be attached to the "Lunar Road-Paving Wagon" to sinter the regolith layer-by-layer [26].

The greater penetration depth relative to solar or laser methods enables stronger surfaces that could be used for industrial loadbearing structures. However, heating from ambient lunar temperatures may require an additional heat source [70]. Further work is needed to optimise microwave sintering systems, particularly to minimise the power per unit volume required [73].

5.4 Direct ink writing

Direct ink writing (DIW) is a type of AM technique that enables the creation of complex structures using a mixture of materials that are formulated as a paste with suitable

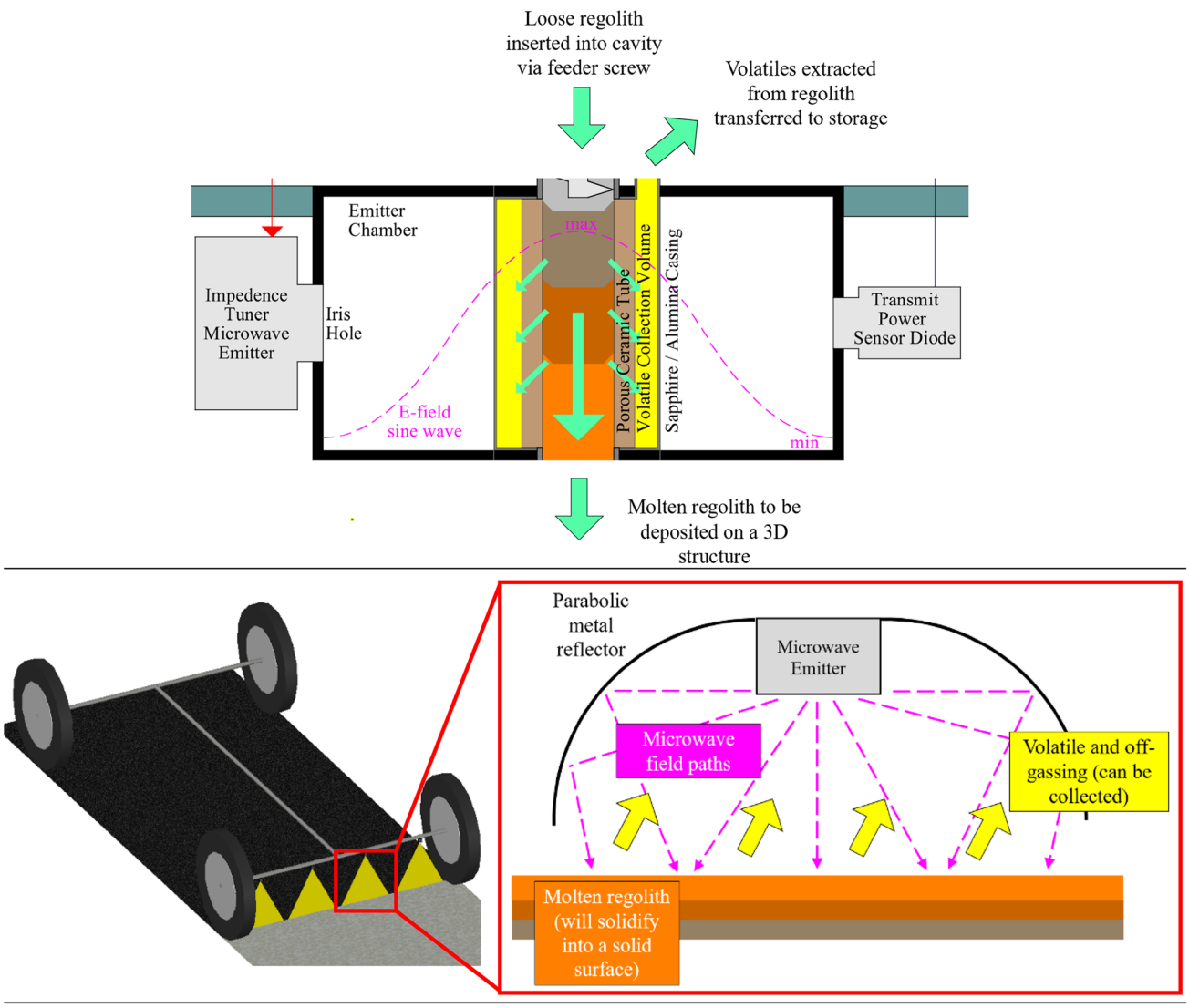

Fig. 9 Top: a diagram of the microwave printhead cavity developed for the ATHLETE autonomous platform, adapted from Howe et al. [111] (With permission from ASCE). Bottom: illustration of "Lunar Road-Paving Wagon" with two sets of magnetrons to effectively sinter lunar soil and a diagram of the mechanism of the microwave emitter adapted from Taylor et al. (Credit: [26])

rheology. Producing high-resolution structures are mainly based on precision motion control systems, often XYZ gantry robot is used, along with printable formulation paste. DIW systems rely heavily on pastes requiring specific rheological properties which comprised of a mixture of powdered materials (lunar soil), liquid solution and certain additives to ensure rheological requirements are met [112].

Soil ink has been developed by Taylor et al. [113] which is comprised of three main parts (i) polylactic acid and glycolic acid polymer, (ii) JSC-1A regolith simulant, and (iii) mixtures of organic solvents. All the components are poured into the ink cartridge and mixed using a mini vortex mixer until the ink reaches ideal viscosity (30–35 Pa.s). Micro-trusses at 10 mm diameter and 20 mm height with 1 mm spacings were printed at 5–6 mm/s extrusion speed and a nozzle diameter of 510 µm. Two-stage heating at 300 °C and 1050–1100 °C was conducted at both air and hydrogen atmospheres. The colour of hydrogen-sintered micro-trusses remained unchanged while air-sintered micro-trusses have more reddish colour due to a change from FeO to Fe_2O_3 as shown in Fig. 10. Compressive strengths of micro-trusses were recorded to range from 1 to 19 MPa and increased relative to increasing density [114].

Jakus et al. [114] developed elastic lunar and martian soil simulants (LRS and MRS) inks synthesised through mixing JSC-1A regolith simulant, PLGA, plasticiser and other various solvents and physically stirred in an open container to achieve printable consistency. 20 mm diameter cylinders and 12 cm long wrenches were successfully printed using LRS and MRS inks (Fig. 11).

Standard sandpaper and silicon carbide paper were used as the print base since adhesion of the first deposited layer is fundamental for the overall printed structure. 3D-printed LRS material properties were heavily reliant on the process parameters, the ensuing porosity, and geometry. Test samples exhibited a Young's moduli in the range of 8–13 MPa and were observed to undergo tensile strains greater than 250% without failure. These 3D-printed structures displayed rubber-like elastic properties under cyclic compression.

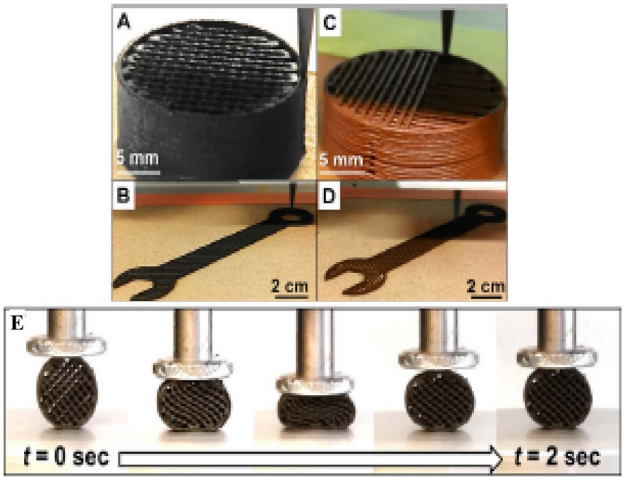

Fig. 11 Images of **A**, **B** LRS and **C**, **D** MRS inks printed into cylinders and wrenches, respectively. **E** Time series photographs of 1 cm-diameter cylinder undergoing compression cycle. (Credit: [114])

Although the DIW technique is very mobile and able to accommodate multiple materials, problems such as storage and inactivation of the organic binder and the µ-gravity extrusion are still prevalent. Despite that, DIW has a high compatibility for printing all types of slurry.

6 Conclusion

This review paper provided a detailed description of the chemical, mineral and morphological compositions of the lunar soil and associated regolith simulants. This information is crucial as regolith is the primary material being considered in constructing lunar habitations as transporting raw materials from Earth is not economically viable. Exploiting local materials is the most likely pragmatic approach to constructing infrastructure on the Moon. Figure 12 displays a summary of the advantages and disadvantages of the sintering methods outlined in this review paper. Key points from

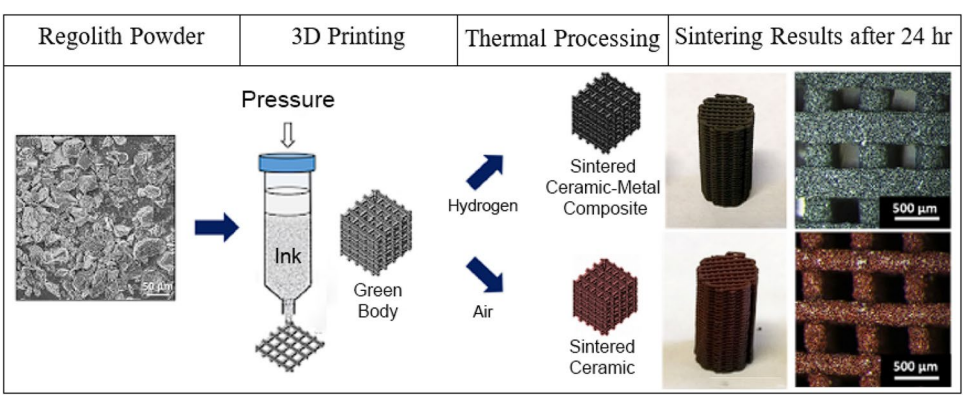

Fig. 10 Schematic diagram of the DIW Printing process and sintering results after 24 h (Credit: [113])

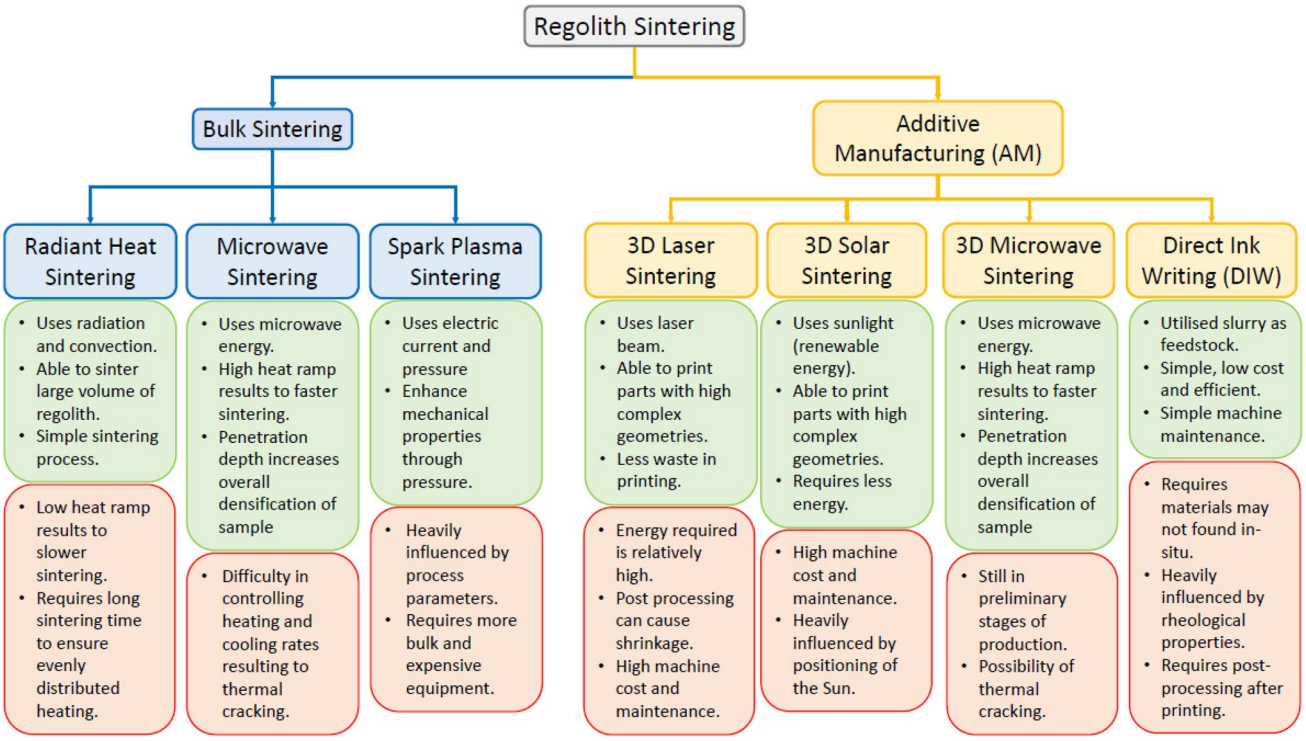

Fig. 12 Summary of the advantages and disadvantages of regolith sintering methods outlined in tree diagram format

this review, and suggestions for further work, are outlined below.

6.1 Bulk sintering

- Sintering studies traditionally used pre-compressed or loose regolith powders at high temperatures to densify the powders into building elements.
- A more recent technique is spark plasma sintering which employs both heating and pressure to achieve higher densification.
- These methods would require large and immobile equipment with slow processing times and high energy consumption.

6.2 Focused sintering

- In contrast to bulk sintering, focused sintering entails fusing successive layers of lunar soil, enabling the in-situ sintering of the lunar surface or as a powder-bed method.
- Laser sintering is energy intensive even for moderately sized parts. Solar sintering applies concentrated energy from the Sun, eliminating the need for external power sources.
- Residual/thermal stresses may limit their strength.

6.3 3D-printing technologies

- Advances in Additive Manufacturing (AM) technologies are critical for the future development of space exploration and construction, as various techniques can be utilised according to specific requirements.
- Applications of 3D printing by solar sintering and Selective Laser Melting (SLM) are capable of printing detailed high-precision parts as tools or building elements.
- Direct Ink Writing (DIW) technology is suited for fabricating larger architectural constructions, though is limited in mechanical strength.
- Transportation of additives and additional materials to accommodate certain 3D-printing techniques, such as DIW, presents an impediment to adoption.

6.4 Recommendations for future work

- There is a need to verify the high-temperature behaviour of lunar simulants. Several studies note phase changes or reactions that affect sintering behaviour, but it is unclear if these reactions are present in lunar soil.
- There is a need to develop fast, low complexity, non-destructive testing methods of lunar soil and sintered materials to ensure the quality of manufactured materials. Nano-indentation or acoustic imaging may be worthy of further investigation to address this challenge.

- Further study of the dynamic properties of sintered materials is needed, including impact resistance and vibration response.
- Sintering methods, particularly focused sintering, need more examination in microgravity and vacuum conditions. Spattering and gassing reactions may be significant for focused sintering and bulk sintering methods respectively.

Data availability The data used to support the findings of this study are included within the article.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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